

Professional Practice, Engineers, and Society

EEE369/ECE369

Department of Electrical and Electronic Engineering
BSRM School of Engineering
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Introduction

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Course Objectives

1. Enable student to understand their **role and responsibilities** as engineering professionals through gaining knowledge of **moral values, philosophies, professional code of ethics and practices**
2. Prepare students to be able to take informed ethical decisions when confronted with problems in the **working environment**
3. Develop students' ability to assess the impact of engineering solutions in the broader **societal and environmental context**
4. Enable students to evaluate the **sustainability** of engineering solutions
5. Improve students' **communication skills** in regard to ethical and professional issues in engineering practice

Course Outcomes (Cos)

Sl.	Course Outcome
CO1	Identify and value the responsibility of the engineers in regard to social, cultural, economic, legal, health, safety and welfare relevant to electrical and electronic engineering solutions and practice.
CO2	Evaluate the sustainability and impact of the electrical and electronic solutions in the broader societal and environmental context
CO3	Resolve competing and complex ethical issues related to the electrical and electronic engineering solutions and professional practices
CO4	Communicate effectively with regard to ethical, professional, societal and environmental issues in electrical and electronic engineering practices and solutions.

Course Outcomes (Cos): CO1

(f) The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (K7)	f1. Demonstrate the contextual knowledge of assessing the societal, health, safety, legal and cultural issues in engineering solutions and professional practice.
	f2. Value the consideration of technical and non-technical issues related to societal, health, safety, legal and culture in developing the solution of complex engineering problem



Course Outcomes (Cos): Explanation of CO1

Engineers have a fundamental responsibility to ensure that their work benefits society while minimizing risks related to social, cultural, economic, legal, health, safety, and welfare aspects. Their decisions impact public safety, environmental sustainability, and ethical integrity, requiring them to adhere to professional codes of ethics and legal regulations.

Example: Ensuring Electrical Safety in a Residential Building

Imagine an electrical engineer designing the wiring system for a new apartment complex. Their responsibilities include:

1. Safety Compliance (Health & Safety Responsibility)

1. Ensuring that electrical circuits follow national safety codes (e.g., NEC, IEC).
2. Installing proper circuit breakers to prevent fire hazards.

2. Economic Responsibility

1. Selecting cost-effective yet reliable materials that balance affordability and durability.
2. Avoiding over-engineering that may increase costs for the residents.

3. Legal & Regulatory Compliance

1. Following building codes and regulations to prevent liability issues.
2. Ensuring that electrical installations are approved by local authorities.

4. Social & Cultural Considerations

1. Designing systems that provide equal access to electricity, considering community needs.
2. Ensuring that electrical infrastructure does not harm historical or cultural sites.

By considering these responsibilities, the engineer protects public safety, ensures sustainability, and upholds ethical standards, demonstrating their professional duty to society.

An electrical engineer is designing the wiring system for a new residential building. The engineer must consider technical and non-technical aspects to ensure safety, compliance, and social responsibility.

1. Assessing Societal, Health, Safety, Legal, and Cultural Issues (f1)

✓ Safety & Health Considerations:

Proper grounding and circuit protection (e.g., circuit breakers, fuses) to prevent electric shocks and fires.

Installing GFCI (Ground Fault Circuit Interrupters) in bathrooms and kitchens to prevent electrocution.

✓ Legal & Regulatory Compliance:

Following national electrical codes (e.g., NEC standards) to ensure legal safety requirements are met.

✓ Cultural & Societal Factors:

Designing an affordable yet safe electrical system, ensuring accessibility for low-income households. Considering local power supply conditions (e.g., unstable voltage levels) and incorporating voltage stabilizers if needed.

An electrical engineer is designing the wiring system for a new residential building. The engineer must consider technical and non-technical aspects to ensure safety, compliance, and social responsibility.

2. Considering Technical & Non-Technical Issues in Electrical Engineering Decisions (f2)

◆ Technical Considerations:

- Choosing high-quality wiring materials (e.g., copper over aluminum for better conductivity and longevity).
- Proper load distribution to avoid overloading circuits and prevent electrical failures.

◆ Non-Technical Considerations:

- Educating homeowners about safe electricity usage (e.g., avoiding overloading power strips/multiplug).
- Ensuring the project stays within budget without compromising safety.

Many people use power strips to connect multiple devices to a single electrical outlet. However, overloading a power strip can cause electrical fires, short circuits, and overheating, posing serious safety hazards.

Course Outcomes (Cos): CO2

(g) Environment and Sustainability: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex electrical and electronic engineering problems in societal and environmental contexts. (K7).	g1. Understand and evaluate the impact of engineering solutions to the environment and society
	g2. Understand the need for sustainable solutions of an engineering problem
	g3. Validate the sustainability of an engineering solution



Course Outcomes (Cos): Explanation of CO2

A company is deciding between **traditional incandescent bulbs** and **LED lighting** for its office buildings. Electrical engineers must evaluate the **environmental impact, sustainability, and long-term benefits** of their choice.

1. Evaluating the Impact of Engineering Solutions on the Environment and Society (g1)

Energy Consumption:

- Incandescent bulbs waste energy as heat, consuming more electricity.
- LED bulbs use 80% less energy, reducing electricity demand and lowering carbon emissions.

Environmental Impact:

- Incandescent bulbs have a shorter lifespan, leading to more waste.
- LEDs last up to 25 times longer, reducing landfill waste and material consumption.

Societal Benefits:

- Lower electricity bills make LEDs cost-effective for households and businesses.
- Reduced energy demand helps prevent power shortages in cities with unstable grids.

Course Outcomes (Cos): Explanation of CO2

2. Understanding the Need for Sustainable Solutions (g2)

◆ Resource Efficiency

- LEDs require less frequent replacements, saving materials and reducing manufacturing pollution.

◆ Renewable Energy Compatibility

- LEDs work well with solar power systems, making them ideal for off-grid rural electrification projects.

3. Validating the Sustainability of the Engineering Solution (g3)

✓ Economic Viability: Although LEDs cost more initially, they pay off through lower energy consumption and reduced maintenance.

✓ Long-Term Environmental Impact: Widespread adoption of LED lighting lowers global energy demand, reducing reliance on fossil fuels.

✓ Scalability: LEDs are used in homes, offices, streetlights, and vehicles, making them a practical and scalable sustainable solution.

Course Outcomes (Cos): CO3

(h) Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of electrical and electronic engineering practice. (K7).	h1. Maintain ethical standards in writing reports, preparing assignments, homework, exams etc.
	h2. Recognize and respect the professional ethics and responsibilities related to engineering practice
	h3. Apply the ethical principles related to the solution of an engineering problem

Ethical Engineering:
Progress with Responsibility



Course Outcomes (Cos): Explanation of CO3

h1. Maintain ethical standards in writing reports, preparing assignments, homework, exams, etc.

In electrical engineering, maintaining ethical standards means presenting accurate information and data.

For example, when preparing a report on the design of a renewable energy system (like a solar power grid), the engineer should avoid exaggerating the system's performance to make it seem more efficient than it actually is. If the calculations show that the system can only provide 80% of the expected energy needs, the engineer must honestly present this result rather than inflating the numbers to make the system appear more viable than it is.

Course Outcomes (Cos): Explanation of CO3

h2. Recognize and respect the professional ethics and responsibilities related to engineering practice

This involves adhering to **safety standards and regulations** in the field. **For example, an electrical engineer working on a power distribution system must follow local electrical codes and standards to ensure safety.** If an engineer is asked to approve a design that doesn't meet these standards (perhaps for cost-saving reasons), they have an ethical responsibility to refuse, as compromising on safety could endanger lives and property.

h3. Apply the ethical principles related to the solution of an engineering problem

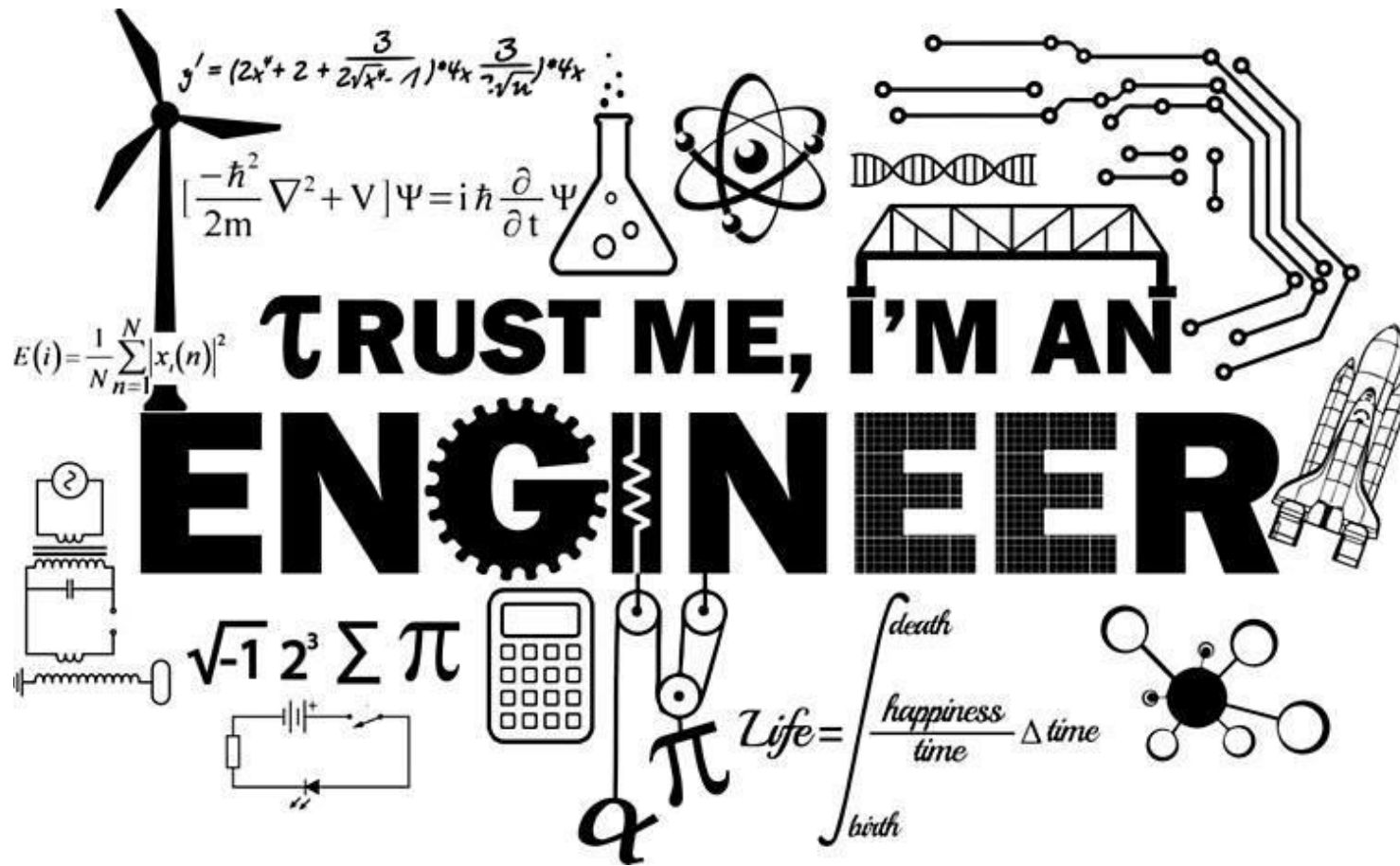
In solving engineering problems, engineers must consider the long-term impacts of their decisions. For example, when designing a new electrical grid, an engineer might be faced with a choice between using cheaper, lower-quality components or higher-quality, more sustainable materials. The ethical choice would be to choose the materials that ensure the system's reliability and safety over the long term, even if it comes at a higher initial cost. This approach prioritizes public safety and environmental sustainability, reflecting the engineer's ethical responsibility.

Content

- ✓ Introduction
- ✓ Ethics and profession
- ✓ Scope of Ethics
- ✓ Code of Ethics for Engineers
- ✓ Case Study

How does it *feel* to
be an engineer?





- Many engineers feel a **sense of accomplishment** in solving **complex problems** and creating innovative solutions that can **impact society positively**.
- There's also a strong sense of **responsibility (ethics)**, especially when working on projects that affect **public safety or the environment** through sustainable technologies.

Why Engineers Feel Accomplished and Responsible

Sense of Accomplishment

Engineers often feel proud and satisfied because they solve real-world problems and create useful solutions.

Examples:

- Designing a **solar power system** that provides electricity to a rural village.
- Developing a **smart traffic control system** that reduces traffic congestion.
- Creating a **water purification system** that supplies clean drinking water.
- Designing a **mobile app** that helps people manage their energy consumption.
- Building a **bridge** that safely connects two communities.

Why it feels rewarding:

- Helps improve people's quality of life.
- Solves difficult technical challenges.
- Contributes to technological progress.
- Creates long-lasting benefits for society.

Why Engineers Feel Accomplished and Responsible

Sense of Responsibility (Engineering Ethics)

Engineers must ensure that their work is safe, reliable, and environmentally friendly because many people depend on their designs.

Examples:

- A civil engineer must design a bridge strong enough to prevent collapse.
- An electrical engineer must ensure power systems operate safely without causing fires.
- A software engineer must protect users' personal data from cyberattacks.
- An environmental engineer must reduce pollution from industrial plants.
- A nuclear engineer must follow strict safety procedures to protect the public and environment.

Why responsibility is important:

- Protects public health and safety.
- Prevents accidents and failures.
- Reduces environmental damage.
- Builds public trust in engineering solutions.
- Supports sustainable development for future generations.

Real-Life Example: The Development of the Three Gorges Dam (China)

The Three Gorges Dam, the world's largest hydroelectric power station, was built on the Yangtze River in China to provide clean energy, flood control, and improved navigation. Engineers involved in the project faced complex challenges related to sustainability, environmental impact, and public safety.



Source: Internet

1. Accomplishment Through Innovation



Solving Energy and Flood Problems:

- The dam generates 22,500 MW of renewable electricity, reducing China's reliance on fossil fuels.
- It significantly controls flooding, protecting millions of people downstream.



Engineering Excellence:

- Engineers had to design a massive structure that could withstand seismic activity and high-water pressure.
- Advanced turbine technology was used to optimize energy output.

2. Ethical and Environmental Responsibility



Social and Environmental Concerns:

- Over 1.3 million people were displaced due to reservoir formation.
- The project altered the local ecosystem, affecting fish populations and biodiversity.



Sustainable Actions Taken:

- Engineers implemented soil erosion control and reforestation programs to mitigate environmental impact.
- Efforts were made to resettle and compensate displaced communities.

The Three Gorges Dam showcases how engineers solve complex problems and innovate to benefit society while also handling ethical dilemmas related to environmental and social concerns. It highlights the balance between technological advancement and ethical responsibility in engineering practice.

Introduction (Summary of the previous slide)

People Generally Think of Engineering as Ethical Professions
and that

We Act in an Ethical Manner

Society Expects Us to Act and Make Decisions in a Way to
Protect the Public

Introduction

Being an Engineer

We are to be Professionals

We are to be Ethical

We are to Protect Society

Then We Should be able to:

Define Profession, Professional and Professionalism

Define Ethics and Ethical Codes

Relate Our Professional Actions to Our Ethical Responsibilities to Society

What is a Profession? What do you think?

- An occupation or vocation requiring **advanced study** in a specialized field.
- The **body of qualified persons** of one specific field.

[American Heritage Dictionary]

Term comes from Latin **professio** meaning to confess or **declare**.

Professionals **declare** themselves to the public as **experts** in a field of study, such as law, medicine, accounting, science and engineering



In a specific field, the body of qualified persons typically consists of individuals who have demonstrated their expertise through formal education, experience, and recognition. These individuals are often nominated or certified by these organizations to ensure they meet the necessary standards for qualifications in that field.

What is a Profession? What do you think?

A Professional is a member of a Profession.
Engineers and other Professionals have:

- ✓ a special responsibility to the public
- ✓ to use their skills and knowledge
- ✓ to ensure safety and promote the well-being of the public

Professionalism means carrying out responsibilities in a way that earns the trust and confidence of the public.



Professional is not a label you give yourself – it's a description you hope others will apply to you.

David Maister

Definition of Profession

A Profession is a disciplined group of individuals who adhere to **ethical standards** and who hold themselves out as, and are accepted by the public as possessing special knowledge and skills in a widely recognized body of learning derived from research, education and training at a high level, and who are prepared to apply this knowledge and exercise these skills in the **interest of others**.

It is inherent in the definition of a **Profession that a code of ethics** governs the activities of each Profession.

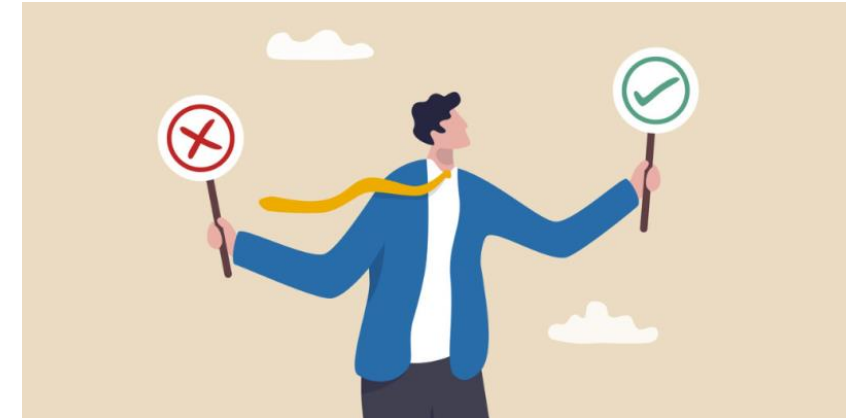
Australian Council of Professions, 2003

Ethics and morals are related concepts, but they have distinct differences

Ethics *vs.* Morals

What are Ethics and Morals?

Aspect	Morals	Ethics
Definition	Personal beliefs about right and wrong.	Systematic rules or principles guiding conduct.
Origin	Derived from culture, religion, and upbringing.	Established by external codes or guidelines. These guidelines are usually set by specific groups or communities. A typical example is the code of ethics followed by doctors, regardless of location or setting. Similarly, lawyers have to follow an established code of ethics.
Application	Subjective; can vary among individuals.	More objective; often codified in guidelines.
Focus	Concerned with personal character and choices.	Concerned with broader societal implications.



— “ —

So far, about morals, I know only that what is **moral is what you feel good after and what is **immoral** is what you feel bad after.**

—Ernest Hemingway

— ” —

Some Examples of Ethics and Morals

Moral	Ethical
An employee believes using company time for personal activities is wrong.	The company's code of conduct prohibits using company resources for personal gain.
A doctor believes assisting patients with euthanasia is immoral.	Some medical guidelines allow for euthanasia, depending on the country or state.
An investor believes that investing in tobacco companies goes against his morals.	The company the investor works for states that investing in tobacco is acceptable because it will benefit the company and their clients.
A student believes that sharing his homework is a moral act because it is generous.	The university strictly prohibits sharing work and classifies it as cheating and, therefore, unethical.
A lawyer tells the judge that his client is guilty because defending him would have gone against his morals.	Lawyers believe that doing this strongly goes against ethical practices.

Professionals use Ethics or Morals?

Professionals

Use **Ethics** to Define

How To Protect

The **Public Good**

Scopes

1. Engineering projects are like social experiments that create new opportunities and risks. Engineers are responsible for making sure their work benefits society, avoids harm, and highlights potential dangers. [When designing a new bridge, engineers must ensure it is safe for users, considering how it will affect the environment, and introduce any risks, like traffic congestion or structural issues]
2. Moral values permeate (fill) all aspects of technological development, and hence ethics and excellence in engineering go together. [When creating a new medical device, engineers must consider patient safety and ethical standards to ensure the device is both effective and safe for users]
3. Personal meaning and commitments matter in engineering ethics, along with principles of responsibility that are stated in codes of ethics and are incumbent (mandatory) on all engineers. [An engineer may feel personally committed to sustainability, and this belief aligns with the code of ethics that requires them to design environmentally friendly buildings]

Scopes

4. Promoting responsible conduct and advocating good works is even more important than punishing wrongdoing.
5. Ethical dilemmas in engineering happen because there are many different moral values that can clash with each other. [An engineer might face a dilemma when designing a product that is cheaper to produce but could harm the environment, creating a conflict between cost-saving and environmental responsibility]
6. Engineering ethics should look at both small-scale (micro) and large-scale (macro) issues, as they are often related. Many ethical challenges are also emerging from the global nature of engineering. [A local engineer might consider how their project impacts the community (micro issue), while also thinking about how it fits into global sustainability efforts (macro issue), such as reducing carbon emissions]
7. Technological development, especially with artificial intelligence, calls for careful optimism—being hopeful but also cautious. [While AI can greatly improve healthcare by providing better diagnoses, we must also be careful about privacy issues and ensuring that the technology is used responsibly]

Canon (fundamental principles or rules) of Ethics

A Statement of Ethical Beliefs and Behaviors
agreed to by a Body of People.

Guides that help determine what is considered ethical behavior.

Guides against which a person can make ethical decisions.

Code of Ethics for Engineers

Each of the Engineering Societies have developed Codes of Ethics. Most of these are based on a model code developed by the Engineering Council for Professional Development (ECPD) in 1947.

Code of Ethics for Association for Computing Machinery (ACM)

ACM developed a Code of Ethics in 1992. The Code states:

“Commitment to ethical professional conduct is expected of every member (voting members, associate members, and student members) of the Association for Computing Machinery (ACM).”

Both the NSPE and the ACM Codes of Ethics are at the end of this section

Code of Ethics for Engineers

We will look at the Code of Ethics developed by the National Society of Professional Engineers (NSPE) as an example. It has four parts:

Preamble

I. The Fundamental Canons

II. The Rules of Practice, and

III. Professional Obligations

(<http://www.nspe.org/Ethics/CodeofEthics/index.html>)

NSPE Code of Ethics for Engineers

Preamble

Engineering is an important and learned profession. As members of this profession, engineers are expected to exhibit the highest standards of **honesty and integrity**. Engineering has a direct and vital impact on the **quality of life** for all people. Accordingly, the services provided by engineers require **honesty, impartiality, fairness, and equity**, and must be dedicated to the protection of the **public health, safety, and welfare**. Engineers must perform under a standard of professional behavior that requires adherence to the highest principles of ethical conduct.

NSPE Code of Ethics for Engineers

I. Fundamental Canons

Engineers, in the fulfillment of their professional duties, shall:

1. **Hold paramount** (top priority) the safety, health and welfare of the public. [An engineer designing a bridge ensures it meets safety standards to protect drivers and pedestrians]
2. Perform services only in areas of their **competence** (expertise). [A civil engineer specializing in road design does not take on a project requiring expertise in structural engineering/A mechanical engineer refrains from taking on an electrical engineering project, recognizing the need for specialized knowledge]
3. Issue public statements only in an **objective** (unbiased) **and truthful** (honest, accurate) manner. [An engineer provides accurate data about a project's environmental impact without exaggeration or misleading information]
4. Act for each employer or client as faithful (trustworthy) agents or trustees. [An engineer discloses all relevant information to a client and prioritizes the client's interests when making recommendations]

NSPE Code of Ethics for Engineers

5. Avoid **deceptive** (misleading) acts. [An engineer does not falsify test results or misrepresent their qualifications to secure a job, An engineer does not manipulate data to make a project appear more cost-effective than it truly is]
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession. [An engineer follows to all relevant laws and regulations, and reports any unethical practices encountered in their work environment]
7. Continue Professional Development Throughout Their Careers. [An engineer regularly attends workshops and courses to stay updated on new technologies and practices in their field]
8. Assist Colleagues and Advise Them of Ethical Issues [A senior engineer advises a junior engineer on how to handle a situation where they are pressured to overlook safety violations]

NSPE Code of Ethics for Engineers

1. Hold Paramount the Safety, Health, and Welfare of the Public

Ethical Practices:

- Always prioritize public safety over personal or organizational gain.
- Follow national and international safety codes (e.g., BNBC, IEEE, ASCE standards).
- Report unsafe practices or defective designs, even if it causes project delays.
- Ensure the design considers environmental protection and disaster resilience.

Real Examples (Ethical):

- ✓ An electrical engineer ensures all wiring in a residential building follows BNBC and IEC safety standards to prevent electrical fires.
- ✓ A civil engineer halts construction when substandard materials are found, even under pressure to continue.

NSPE Code of Ethics for Engineers

1. Hold Paramount the Safety, Health, and Welfare of the Public

Unethical Practices:

- Ignoring known safety issues to save cost or meet deadlines.
- Approving defective materials or low-quality work to satisfy contractors or clients.
- Hiding potential hazards from the public or authorities.

Real Examples (Unethical):

✗ The collapse of the Rana Plaza building in Bangladesh (2013) — engineers approved illegal structural modifications without ensuring safety compliance.

✗ An engineer ignores reports of overheating in electrical panels to keep production running.

NSPE Code of Ethics for Engineers

2. Perform Services Only in Areas of Their Competence

Ethical Practices:

- Accept only tasks that match your education, training, and experience.
- Seek expert collaboration when work involves unfamiliar disciplines.
- Continuously update knowledge through professional development and certifications.

Real Examples (Ethical):

-  A mechanical engineer consults an electrical engineer for designing control systems in an automated plant.
-  A civil engineer declines to design a nuclear containment structure, as it requires specialized nuclear engineering expertise.

NSPE Code of Ethics for Engineers

2. Perform Services Only in Areas of Their Competence

Unethical Practices:

- Taking projects outside one's expertise for financial or reputational gain.
- Signing off on another engineer's specialized design without full understanding.
- Failing to disclose lack of competence to clients or employers.

Real Examples (Unethical):

 A mechanical engineer attempts to design a high-voltage substation without electrical background, leading to unsafe layouts.

 A software engineer claims expertise in cybersecurity without proper training, resulting in a data breach.

NSPE Code of Ethics for Engineers

3. Issue Public Statements Only in an Objective and Truthful Manner

Ethical Practices:

- Present data, findings, and opinions based on verified facts and sound engineering judgment.
- Avoid exaggeration, manipulation, or omission of important details.
- Clearly state limitations, assumptions, or uncertainties in reports.

Real Examples (Ethical):

-  An environmental engineer honestly reports that a new factory design meets 80% of pollution control targets, recommending additional filters.
-  A project manager discloses both benefits and risks of installing a new wind turbine technology in a feasibility study.

NSPE Code of Ethics for Engineers

3. Issue Public Statements Only in an Objective and Truthful Manner

Unethical Practices:

- Making biased statements to favor a company or product.
- Hiding or altering negative results to obtain funding or approval.
- Using technical language to mislead non-expert clients or the public.

Real Examples (Unethical):

✗ The Volkswagen emission scandal — engineers manipulated emission data to falsely meet environmental standards.

✗ A consultant claims that a bridge design is “fully earthquake-proof” without proper testing or analysis.

NSPE Code of Ethics for Engineers

4. Act for Each Employer or Client as Faithful Agents or Trustees

Ethical Practices:

- Maintain loyalty, honesty, and confidentiality with clients and employers.
- Avoid conflicts of interest that could influence professional judgment.
- Disclose any potential financial or personal interests before accepting a project.
- Protect proprietary and confidential information of the client.

Real Examples (Ethical):

-  An engineer refuses to share confidential project data of one client with another competing firm.
-  A project consultant discloses that he owns shares in a supplier company before recommending its products.

NSPE Code of Ethics for Engineers

4. Act for Each Employer or Client as Faithful Agents or Trustees

Unethical Practices:

- Accepting bribes or gifts to influence project decisions.
- Using a client's confidential design in another project without permission.
- Working for competing clients on similar projects without full disclosure.

Real Examples (Unethical):

- ✗ An engineer awards a contract to a supplier in exchange for personal commission.
- ✗ A design firm copies a client's proprietary turbine blade design for another manufacturer.

NSPE Code of Ethics for Engineers

5. Avoid Deceptive Acts

Ethical Practices:

- Always be honest and transparent in professional activities.
- Do not falsify or misrepresent qualifications, experience, or project outcomes.
- Report any form of bribery, fraud, or corruption observed in projects.
- Ensure advertisements, reports, and proposals are factual and not misleading.
- Avoid plagiarism in technical or academic works.

Real Examples (Ethical):

- ✓ An engineer clearly states in a proposal that their team has “3 years of experience” in solar plant design — not more.
- ✓ A contractor informs the client about a supplier’s attempt to offer a bribe to secure a material supply deal.
- ✓ An academic researcher properly cites others’ work in a published paper.

NSPE Code of Ethics for Engineers

5. Avoid Deceptive Acts

Unethical Practices:

- Making false claims about qualifications or project success.
- Submitting forged certificates or manipulating test data.
- Taking credit for another engineer's work or ideas.
- Engaging in bribery or corruption to win contracts.

Real Examples (Unethical):

- ✗ An engineer alters load test results of a bridge to make it appear safe for use.
- ✗ A company claims to have ISO certification that it doesn't actually possess.
- ✗ A researcher copies diagrams or formulas from another's thesis without citation.
- ✗ A contractor pays a procurement officer to win a government project.

NSPE Code of Ethics for Engineers

6. Conduct Themselves Honorably, Responsibly, Ethically, and Lawfully to Enhance the Honor, Reputation, and Usefulness of the Profession

Ethical Practices:

- Behave professionally and respectfully in all work-related activities.
- Abide by laws, codes of ethics, and standards of professional conduct.
- Support and mentor younger engineers with honesty and encouragement.
- Promote sustainable and responsible engineering practices.
- Report unethical or illegal conduct by peers when observed.

Real Examples (Ethical):

- ✓ An experienced engineer mentors junior staff on safe design standards and ethical decision-making.
- ✓ A project manager refuses to sign off on substandard work even under political pressure.
- ✓ A firm adopts sustainable design principles to reduce carbon emissions.
- ✓ An engineer reports a colleague's falsified certification to the licensing board.

NSPE Code of Ethics for Engineers

6. Conduct Themselves Honorably, Responsibly, Ethically, and Lawfully to Enhance the Honor, Reputation, and Usefulness of the Profession

Unethical Practices:

- Damaging the reputation of the profession through misconduct or illegal acts.
- Exploiting junior engineers or taking credit for their work.
- Ignoring professional misconduct of peers for personal convenience.
- Violating environmental laws or human rights in project execution.

Real Examples (Unethical):

- ✗ A senior engineer falsifies academic credentials and damages public trust in the profession.
- ✗ A firm illegally dumps toxic waste to cut disposal costs, harming communities.
- ✗ A manager takes sole credit for a team's innovation in a completed project.
- ✗ Engineers remain silent about safety violations to avoid conflict with management.

NSPE Code of Ethics for Engineers

7. Continue Professional Development Throughout Their Careers

Ethical Practices:

- Engage in lifelong learning by attending workshops, seminars, and professional training.
- Stay updated with emerging technologies, tools, and standards in the field.
- Encourage and support professional development of subordinates and peers.
- Pursue professional certifications and higher education to maintain technical competence.
- Participate in professional societies (e.g., IEB, IEEE, ASME) for knowledge sharing.

Real Examples (Ethical):

- ✓ An electrical engineer regularly attends IEEE training on renewable energy integration.
- ✓ A power engineer updates their skills by learning new analysis tools such as ETAP, PowerFactory
- ✓ A company sponsors its engineers to attend continuing education programs to maintain their licenses.
- ✓ A mechanical engineer takes a course on smart manufacturing technologies to stay competitive.

NSPE Code of Ethics for Engineers

7. Continue Professional Development Throughout Their Careers

Unethical Practices:

- Ignoring new developments or relying only on outdated methods learned years ago.
- Refusing training opportunities due to complacency or overconfidence.
- Misrepresenting one's knowledge as "up-to-date" without actual effort.
- Discouraging junior engineers from pursuing further education to save time or costs.

Real Examples (Unethical):

✗ A senior engineer continues using outdated electrical safety codes, leading to non-compliance with modern standards.

✗ A project manager refuses to adopt new renewable grid modeling software, causing project inefficiencies.

✗ A supervisor denies staff permission to attend skill development workshops to reduce expenses.

NSPE Code of Ethics for Engineers

8. Assist Colleagues and Advise Them of Ethical Issues

Ethical Practices:

- Support colleagues in ethical decision-making and professional growth.
- Provide honest, constructive feedback when unethical behavior is observed.
- Mentor junior engineers on how to handle professional dilemmas (e.g., pressure from management, data manipulation).
- Promote a culture of integrity, collaboration, and mutual respect in the workplace.
- Report serious ethical violations to appropriate authorities when necessary.

Real Examples (Ethical):

- ✓ A senior engineer helps a junior who is pressured to approve unsafe construction materials to report the issue confidentially.
- ✓ A manager holds ethics training sessions to educate staff about company policy and professional codes.
- ✓ A colleague discreetly advises another to correct a technical report that contained misleading cost estimates.
- ✓ An engineer encourages open discussions on ethics during project meetings.

NSPE Code of Ethics for Engineers

8. Assist Colleagues and Advise Them of Ethical Issues

- Ignoring or covering up unethical actions by colleagues.
- Encouraging shortcuts or unsafe practices to meet deadlines.
- Discouraging peers from reporting violations or “whistleblowing.”
- Using mentorship or seniority to manipulate or exploit junior engineers.

Real Examples (Unethical):

- ✗ A team leader tells a junior engineer to “just sign off” on an incomplete safety inspection.
- ✗ A senior engineer knows a peer falsified data but remains silent to avoid conflict.
- ✗ A manager pressures employees not to report safety violations to keep the company’s image clean.
- ✗ A senior engineer claims credit for a junior’s innovative solution without acknowledgment.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)
 - a. If engineers' judgment is overruled under circumstances that endanger life or property, they shall notify their employer or client and such other authority as may be appropriate. [If an electrical engineer identifies that the design of an electrical system does not comply with safety standards—such as inadequate circuit protection that could lead to overheating or fire—but is instructed by a project manager to proceed with the design, the engineer should first inform the project manager about the potential dangers. If the project manager insists on ignoring the issue, the engineer should then notify relevant authorities, such as the local electrical inspection office, to prevent any potential hazards to life or property]
 - b. Engineers shall approve only those engineering documents that are in conformity with applicable standards. [An electrical engineer reviews the plans for a new building's wiring system. They notice that the proposed designs do not comply with the National Electrical Code (NEC) standards regarding grounding and circuit protection. Before signing off on the documents, the engineer requires the design team to revise the plans to meet these safety standards, ensuring that the final approved documents conform to all applicable regulations]

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)
- c. Engineers shall not reveal facts, data, or information without the prior consent of the client or employer except as authorized or required by law or this Code. [An electrical engineer works on a project for a corporate client that includes sensitive information about their data center's security systems. During a discussion with a vendor, the engineer is asked for details about the project specifications. The engineer refuses to share any information without the client's prior consent, respecting the confidentiality agreement, and only shares what is necessary under the terms of the project contract]
- d. Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe are engaged in fraudulent or dishonest enterprise. [An electrical engineer is approached by a contractor to collaborate on a new project. During discussions, the engineer discovers that the contractor has a history of cutting corners and using substandard materials to save costs. Believing that this could lead to unsafe practices, the engineer declines to associate with the contractor and refuses to put their name on any documents or promotional materials related to the project, ensuring their professional integrity remains intact]

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)
- e. Engineers shall not aid or abet the unlawful practice of engineering by a person or firm. [An electrical engineer is approached by a friend who wants to start an electrical contracting business but lacks the necessary licenses. The friend asks the engineer to sign off on projects to make it appear that the work is being done by a licensed professional. Recognizing this as an unlawful practice, the engineer refuses to participate, explaining the legal implications and emphasizing the importance of following licensing regulations in the industry]
- f. Engineers having knowledge of any alleged violation of this Code shall report there on to appropriate professional bodies and, when relevant, also to public authorities, and cooperate with the proper authorities in furnishing such information or assistance as may be required. [An electrical engineer learns that a colleague has been consistently using outdated safety practices and failing to follow the National Electrical Code (NEC) in their projects. Concerned about the potential hazards this poses, the engineer reports the violations to the appropriate professional engineering board. Additionally, if the situation poses immediate danger to public safety, the engineer also notifies local building authorities to ensure that necessary actions are taken to address the issue]

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Explanation of a, b. c. d. e and f under Canon I is found from the next slides where ethical & unethical practices of each rule of practices are provided.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule a: If engineers' judgment is overruled under circumstances that endanger life or property, they shall notify their employer or client and such other authority as may be appropriate.

Ethical Practices:

- Inform supervisors or clients immediately if a decision endangers public safety.
- Document safety concerns in writing to create a record of professional responsibility.
- Escalate the issue to higher authorities (such as regulatory boards or safety inspectors) if management ignores safety warnings.
- Refuse to approve or continue work that violates safety standards or legal requirements.

Real-Life Ethical Examples:

 An electrical engineer discovers undersized cables in a design that may cause overheating. They warn the project manager and, when ignored, report it to the building safety inspector.

 A civil engineer objects to using substandard cement for a bridge project and informs the Department of Public Works after being overruled by management.

 A mechanical engineer halts testing of a pressure vessel when noticing unsafe pressure levels, despite production deadlines.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule a: If engineers' judgment is overruled under circumstances that endanger life or property, they shall notify their employer or client and such other authority as may be appropriate.

Unethical Practices:

- Remaining silent when management overrules a safety-related judgment.
- Ignoring or hiding design flaws to avoid project delays or conflict with supervisors.
- Signing off on a design known to pose risks to human life or property.
- Failing to notify the proper authority about unsafe decisions or practices.

Real-Life Unethical Examples:

-  A project engineer proceeds with a design after discovering an inadequate grounding system but fails to report it, leading to an electrical fire.
-  An architect ignores warnings about faulty structural columns to keep the project on schedule, resulting in a collapse.
-  Engineers in the Boeing 737 MAX case failed to fully disclose software safety issues to regulatory authorities, leading to fatal accidents.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule b: Engineers shall approve only those engineering documents that are in conformity with applicable standards.

Ethical Practices:

- Carefully review all engineering drawings, plans, and specifications before approval.
- Ensure compliance with applicable codes and standards (e.g., BNBC, NEC, IEC, ASME, ISO).
- Require corrections or revisions for any nonconforming documents before signing off.
- Verify that materials, systems, and designs meet performance and safety requirements.
- Keep records of all approved documents and justifications for decisions.

Real-Life Ethical Examples:

-  An electrical engineer reviewing a building's wiring design refuses to approve it until grounding and circuit breaker protections meet NEC and BNBC standards.
-  A civil engineer detects that a proposed drainage design doesn't meet flood protection standards and requests redesign before signing.
-  A structural engineer checks that load calculations follow international seismic design codes before final approval.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule b: Engineers shall approve only those engineering documents that are in conformity with applicable standards.

Unethical Practices:

- Approving documents without proper review or technical verification.
- Ignoring noncompliance with safety or design standards to meet deadlines.
- Allowing substandard materials or design shortcuts to pass approval for financial gain.
- Delegating document approval to unqualified staff or signing without personal review.

Real-Life Unethical Examples:

-  An engineer signs off on electrical blueprints without verifying compliance with NEC — later causing overheating and short circuits.
-  A building consultant approves a noncompliant design to avoid conflict with the client, resulting in structural failure.
-  A contractor pressures an engineer to approve substandard steel quality — the engineer agrees, leading to a collapse during construction.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule c: Engineers shall not reveal facts, data, or information without the prior consent of the client or employer, except as authorized or required by law or this Code.

Ethical Practices:

- Maintain strict confidentiality regarding clients' or employers' sensitive data.
- Share project-related information only with authorized personnel or parties involved in the contract.
- Obtain written consent from the client before disclosing confidential information to third parties.
- Disclose information only when legally required (e.g., in case of safety violations, environmental hazards, or court orders).
- Secure digital and printed project data using passwords, encryption, or controlled access.

Real-Life Ethical Examples:

-  An electrical engineer working on a corporate data center refuses to disclose server layout details to a supplier without client approval.
-  A consultant shares only the necessary, approved data with the contractor to ensure project confidentiality.
-  A civil engineer informs environmental authorities about illegal waste dumping only after verifying that disclosure is required by law to protect public health.
-  A renewable energy engineer stores confidential solar farm performance data in an encrypted format to prevent unauthorized access.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule c: Engineers shall not reveal facts, data, or information without the prior consent of the client or employer, except as authorized or required by law or this Code.

Unethical Practices:

- Disclosing confidential information without permission.
- Sharing client data with vendors or competitors for personal benefit.
- Using a former employer's confidential design or pricing strategy for another company.
- Discussing sensitive project details publicly, such as on social media or at conferences, without authorization.

Real-Life Unethical Examples:

-  An engineer leaks blueprints of a high-security facility to a supplier to win future contracts.
-  A consultant discusses a client's financial data in a conference presentation without obtaining consent.
-  A former employee uses a previous company's proprietary design for a new employer.
-  An engineer shares restricted project details with a journalist, breaching a confidentiality agreement.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule d: Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe are engaged in fraudulent or dishonest enterprise.

Ethical Practices:

- Conduct due diligence before associating with individuals, firms, or organizations.
- Refuse to endorse or sign documents for dishonest or fraudulent entities.
- Withdraw professional association immediately upon discovering unethical or illegal practices.
- Report known fraudulent activities to professional boards or authorities.
- Protect one's professional reputation by avoiding companies with histories of corruption, substandard work, or misrepresentation.

Real-Life Ethical Examples:

-  An engineer refuses to partner with a contractor known for falsifying safety test results.
-  A design consultant withdraws from a joint venture after discovering financial fraud within the partner company.
-  A civil engineer denies the use of their license seal on documents submitted by an unlicensed firm.
-  A professional engineer reports a company that sells fake certificates of compliance for profit.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule d: Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe are engaged in fraudulent or dishonest enterprise.

Unethical Practices:

- Allowing one's name, license, or credentials to be used by unqualified or dishonest persons or firms.
- Partnering with companies known for fraudulent practices to gain business benefits.
- Overlooking corruption, bribery, or substandard practices for personal gain.
- Acting as a “rubber stamp” engineer by approving documents not prepared or verified by oneself.

Real-Life Unethical Examples:

-  An engineer lends their license to a contractor to approve illegal construction work.
-  A professional knowingly collaborates with a firm involved in bribery to secure government tenders.
-  An engineer signs documents prepared by non-qualified technicians for money.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule e: Engineers shall not aid or abet the unlawful practice of engineering by a person or firm.

Ethical Practices

- Refusing to sign or approve engineering documents for unlicensed individuals or firms.
 - *Example:* An electrical engineer declines to sign design documents for a contractor who doesn't hold a professional engineering license.
- Explaining to others the legal and professional implications of performing engineering work without authorization.
 - *Example:* The engineer educates a friend about the importance of professional licensing before taking on engineering work.
- Reporting unauthorized or fraudulent engineering practices to the licensing board or relevant authority.
 - *Example:* A licensed engineer reports a company falsely advertising "engineering consulting" services without a registered engineer.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule e: Engineers shall not aid or abet the unlawful practice of engineering by a person or firm.

Unethical Practices

- Allowing one's name, seal, or signature to be used on engineering documents prepared by unqualified or unlicensed individuals.
 - *Example:* An engineer signs off on a project designed entirely by an unlicensed technician to help them get approval faster.
- Participating in or covering up unlicensed engineering work for profit or friendship.
 - *Example:* Approving blueprints for a friend's contracting company despite knowing they are not legally qualified.
- Ignoring regulatory or licensing violations within a firm.
 - *Example:* Turning a blind eye to a coworker's unauthorized stamp usage on public documents.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule f: Engineers having knowledge of any alleged violation of this Code shall report thereon to appropriate professional bodies and, when relevant, also to public authorities.

Ethical Practices

- Reporting unethical, unsafe, or unlawful engineering practices to professional bodies or government authorities.
 - *Example:* An electrical engineer reports a colleague who consistently ignores safety codes in power system installations.
- Providing full cooperation to authorities during investigations of ethical or safety violations.
 - *Example:* When a professional board investigates a faulty bridge design, the engineer shares technical evidence and design data transparently.
- Acting promptly when there is an imminent threat to public safety.
 - *Example:* An engineer immediately alerts municipal authorities when discovering a construction site using substandard electrical materials.

NSPE Code of Ethics for Engineers

II. Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public. (Canon 1)

Rule f: Engineers having knowledge of any alleged violation of this Code shall report thereon to appropriate professional bodies and, when relevant, also to public authorities.

Unethical Practices

- Ignoring known ethical or legal violations within a project or organization.
 - *Example:* An engineer notices falsified test results but stays silent to avoid conflict with superiors.
- Concealing information from authorities investigating safety or code violations.
 - *Example:* Refusing to provide data to regulators during an investigation of a failed electrical installation.
- Protecting colleagues engaged in misconduct out of loyalty or friendship.
 - *Example:* Choosing not to report a senior engineer who bypasses safety tests to meet project deadlines.

Find the all rules of practice under different Canons

(<http://www.nspe.org/Ethics/CodeofEthics/index.html>)

Case Studies

- Split into four groups and each group review one of the cases.
- **Select the Canon** that you believe to be most appropriate for the question(s) asked.
- Then find the corresponding “**Rule of Practice**” and read the clauses.
- Be ready to briefly describe the case and what “**Rules**” you believe are significant.
- These examples are from the **NSPE Magazine**.

Case Studies

FACTS: Case 1

Engineer A is involved with Buildings and Grounds at an Army base. He has been asked to **certify** that certain storage rooms and arms storage racks are in compliance with extensive **Army regulations**.

Question:

Would it be appropriate for Engineer A to certify as a qualified engineer the arms storage rooms and storage racks as requested by the Army?

Case Studies: **Explanation of the Ques**

Certification means that **Engineer A is verifying the design, safety, and compliance** of the storage rooms and racks based on Army regulations. However, if Engineer A **does not have expertise in this area** (such as security structures or weapons storage requirements), they should **not certify it**.

Case Studies

References of Case 1

1. **Section II.1.** - Code of Ethics: Engineers shall hold paramount the safety, health and welfare of the public.
2. **Section II.2.a.** - Code of Ethics: Engineers shall undertake assignments only when qualified by education or experience in the specific technical fields involved.
3. **Section II.2.b.** - Code of Ethics: Engineers shall not affix their signatures to any plans or documents dealing with subject matter in which they lack competence, nor to any plan or document not prepared under their direction and control.

Case Studies

Discussion on Case Study 1

- 1) "... Licensed engineers must make all efforts to perform professional services solely within their **area of competence** and not be unduly influenced either by employer or by client pressures that could cause grave danger to the **public health and safety**."
- 2) "... even if **Engineer A had the ethical competency to perform the services required**, it would not have been ethically proper for Engineer A to "certify" compliance with the military regulations as requested. "

Conclusion:

It **would not be ethical** for Engineer A to certify as a qualified engineer the arms storage rooms and arms storage racks as requested by the Army official.

Case Studies: Summary

Why?

1. **Ethical Responsibility** – Engineers should only approve things they are **competent** in.
2. **Legal Risk** – If something goes wrong (e.g., security failure, theft, or accident), Engineer A **could be held responsible**.
3. **Professional Standards** – Engineering codes of ethics require that certifications be made only within one's field of expertise.

What Should Engineer A Do?

- **Seek help from a qualified expert** (e.g., a structural or security engineer).
- **Decline the request if they are not qualified** in that area.

Case Studies

Case Study 2

FACTS: A structural engineer, Ms. Goode is **hired by a newspaper** to visit a bridge site which has had a series of serious delays and cost increases. The engineer performs a one-day visual inspection. She reports some potential problems and proposes some additional testing. The paper **writes a series of articles alleging major safety problems which will delay the bridge opening**. She responds that **her report was not specific and discussed only potential problems with the bridge**.

Question: Was it ethical for Ms. Goode to agree to perform an investigation for the newspaper in the manner stated?

Case Studies

Case Study 2 (References):

1. Code of Ethics Section II.3.a. "Engineers shall be objective and truthful in professional reports, statements or testimony. They shall include all relevant and pertinent information in such reports, statements or testimony."
2. Section II.3.b. "Engineers may express publicly a professional opinion on technical subjects only when that opinion is founded upon adequate knowledge of the facts and competence in the subject matter."
3. Section II.3.c. "Engineers shall issue no statements, criticisms or arguments on technical matters which are inspired or paid for by interested parties, unless they have prefaced their comments by explicitly identifying the interested parties on whose behalf they are speaking, and by revealing the existence of any interest the engineers may have in the matters."

Case Studies

Discussion on Case Study 2

- Engineers should “become active and involved in matters concerning the well-being of the public. Moreover, the NSPE Code of Ethics makes clear that engineers should “seek opportunities to be of constructive service in civic affairs and work for the advancement of the safety, health and well-being of their community.” (Section III.2.a.) “
- “...the engineer has an obligation to the public as well as to the profession to protect the integrity of her professional opinions and the manner in which those opinions are disseminated to the public.”

Conclusion: It was **not unethical** for structural Engineer to agree to perform an investigation **for the newspaper in the manner** stated but Engineer has an obligation to require the newspaper to state in the article that Engineer had been retained for a fee by the newspaper to provide her professional opinion concerning the safety of the bridge.

References

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4. Fleddermann, Fleddermann Charles Byrns - Engineering ethics-Pearson_ Prentice Hall (2011_2012)
5. Different internet sources

The cases and examples in this lecture are mainly taken from different internet sources as well as generated by Google AI when needed to make the lecture better understandable for students. The cases and examples are cross-checked before placing in this lecture.

Citation idea is taken from <https://guides.lib.wayne.edu/c.php?g=1368920&p=10228374>

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